

MBSD Assignment #4 A.Y. 2024/25

Purposes

- Integrate the one-pedal controller into a simulated Arduino Uno¹ microcontroller (μ C), resorting to SimulIDE².
- Interact with the μ C through its digital and analog interfaces.

Instructions

For instructions on how to use SimulIDE and the Platform Support Packages, follow the instructions **provided by Prof. Violante in the lectures.**

The delivery shall contain:

- The controller Simulink model is used to generate the firmware binary file (plus all the accompanying files needed to make it possible to generate the code again, like .m files containing initializations)
- The firmware binary file to be loaded into the simulated Arduino in SimulIDE
- The SimulIDE project file.
- The PDF or Microsoft Word version of the report.

Is available an example based on a Tank level controller in the folder

The deliverable has to be provided as a . ZIP file up to June 8th at 23:59. It shall also contain a brief report explaining the integration process using the following template. It is sufficient that only one of the group members uploads it.

¹ Arduino Uno Board Anatomy <https://docs.arduino.cc/tutorials/uno-rev3/BoardAnatomy>, last visited on 14/05/2025.

² SimulIDE, <https://www.simulide.com/p/home.html>, last visited on 14/05/2025.

Model-Based Software Design, A.Y. 2024/25

Assignment 4 Report

Components of the working group (max 2 people)

- Mahdy Mokdad 346967
- Mohammad Nour Edeen 344035

I/O interfaces

[Please describe the I/O interfaces to interact with the one pedal controller through the electrical interfaces of the Arduino]

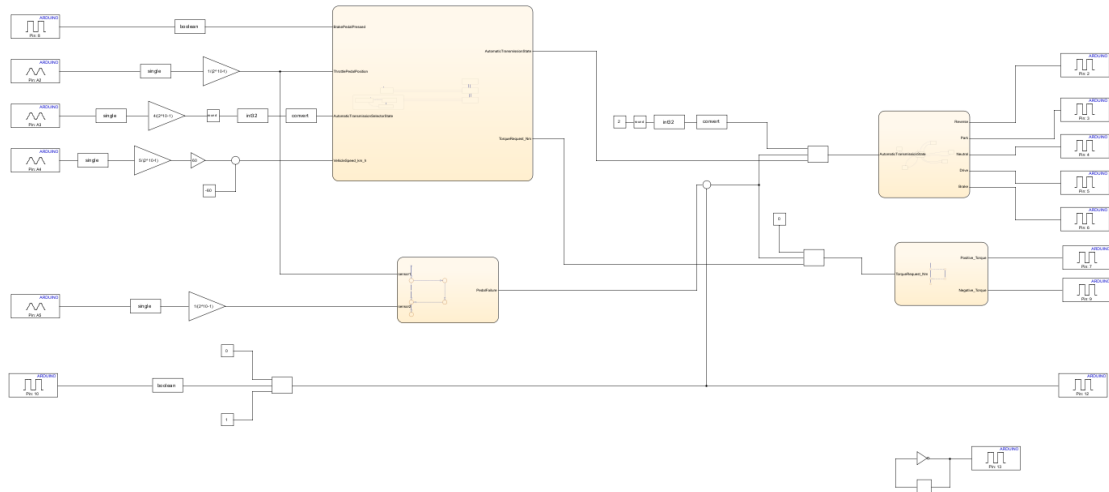
[Screenshot of the controller model ready for the code generation]

Name	Unit	Type ³	Conversion formulas	Min ⁴	Max
Thriddle Pedal Position	N/A	single	$1/(2^{10}-1)$	0	1V
AutomaticTransmissionSelectorState	N/A	Int32	$4/(2^{10}-1)$	0	5V
VehicleSpeed_kmh	Km/h	single	$5/(2^{10}-1)$	0	5V
BrakePedalPressed	N/A	Boolean	N/A	N/A	N/A
AutomaticTransmissionState	N/A	State			
TorqueRequest_Nm	Nm	Int32			

³ Digital Input (DI), Digital Output (DO), Analog Input (AI).

For AIs, provide the conversion formula from input voltage to the measurement unit data (indicating also how to perform the conversion from the raw reading of the ADC).

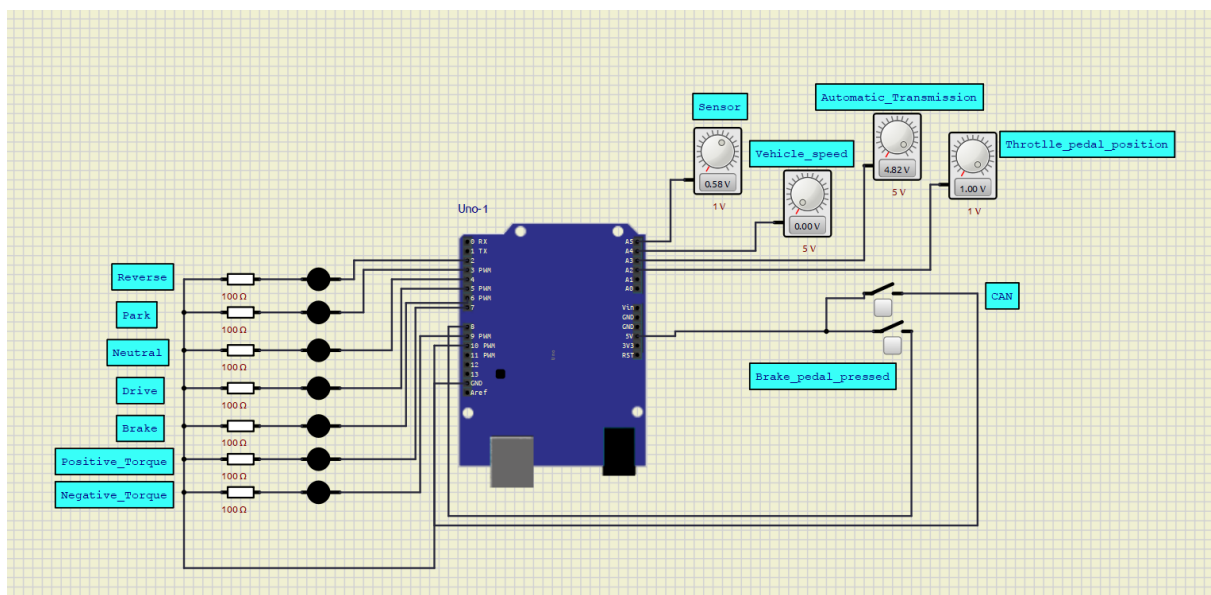
⁴ The Min/Max values that can be handled due to the conversion formula shall be expressed in the measurement unit specified in the Unit column.



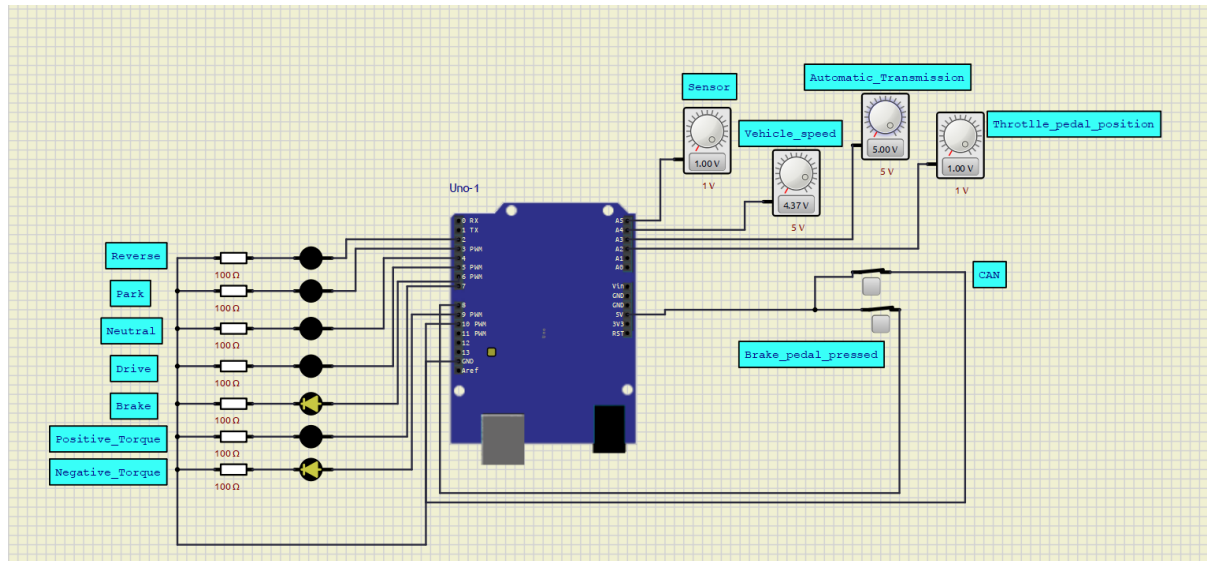
Code generation for Arduino

To generate the Arduino firmware, first the Arduino controller has been designed in order to meet the safety requirements to be implement into the actual board on SIMULide. After tha the model settings are configured in order to meet our preferences in terms of code generation and hardware implementaions. After that, the embedded code and Simulink coder toolboxes have been used in order to build up the model. One of the generated files, alongside the c code of our model is a .hex file that replicates our controller on an Arduino controller. Such file is uploaded to be excuted later.

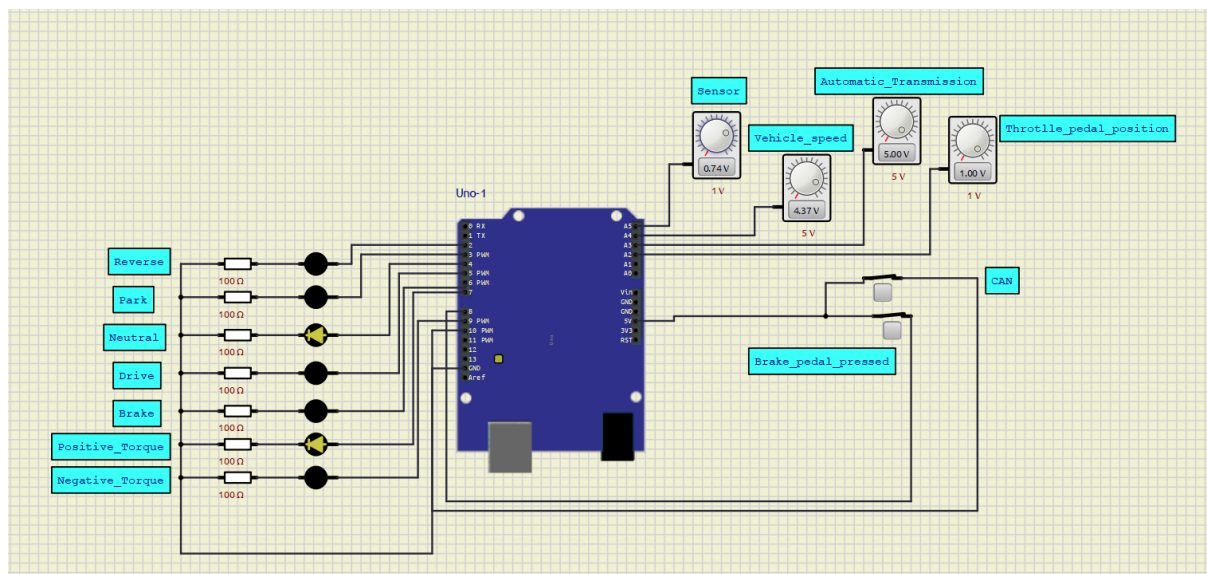
Harness



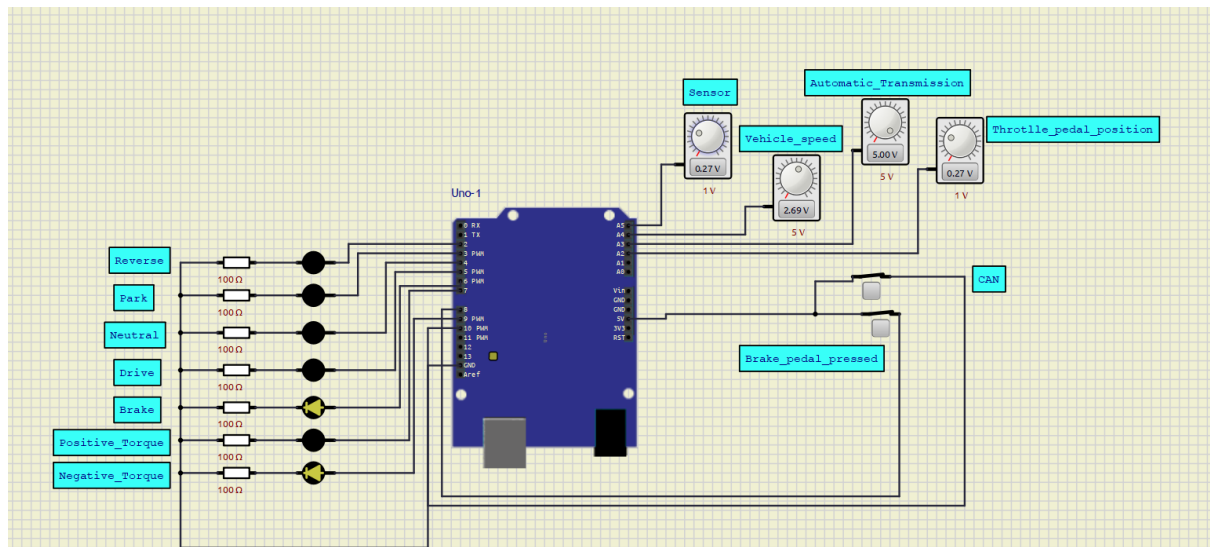
Test stimuli



For this configuration 5 volts have been fed for the Automatic-transmission and consistent positions for the 2 sensors have been detected, thus a brake and a positive torque will appear on the output.

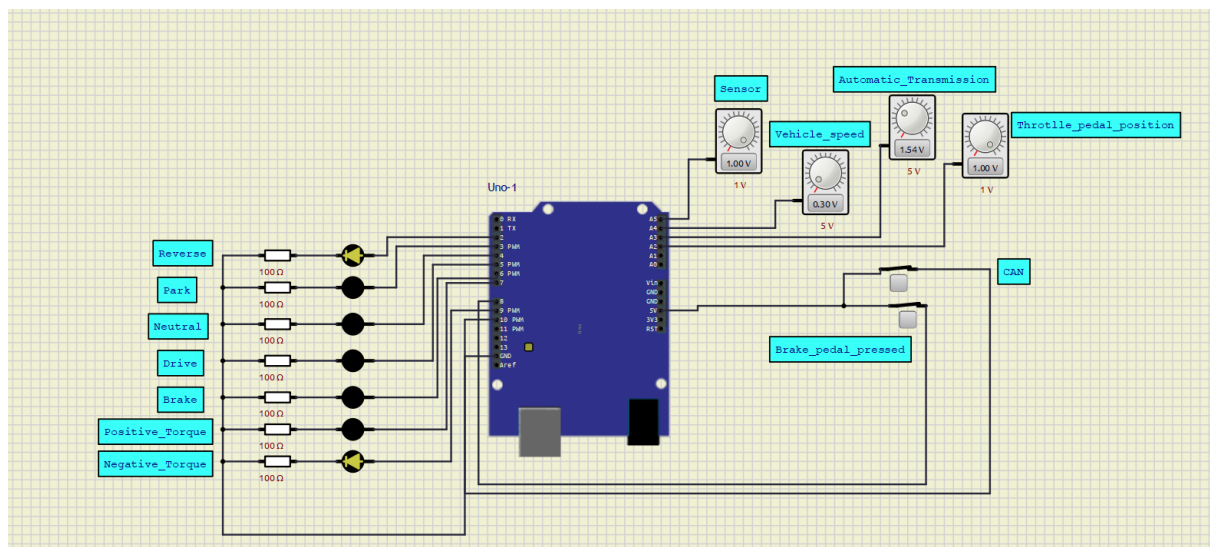


Once one sensor records a different value the transmission state is turned to neutral. (The torque will be zero in this case but is taken to be positive).



For this case the pedal position is less than 0.3 and thus a negative torque will be selected at the output when the transmission is at brake state.

Note that the brake pedal and the CAN network are indicated with a button and they should be pressed to transmission from state to state and to ensure safe state respectively.



And finally for the case of transmission state 1 (which will be from 1 to 2 V in this case taking into consideration the rounding effect). The torque would be negative accordingly.